

Gavin Perrell

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Professional Summary

Math graduate with extensive experience implementing math models into software, validating model performance, and presenting detailed documentation to teams. Extensive experience in **Python**, **R**, **MATLAB**, and **Excel**. Looking to apply coursework, including advanced statistical programming and numerical methods for PDEs, to the analysis of stochastic markets.

Education

South Dakota State University, Brookings, SD

M.S. Mathematics, Aug. 2026

GPA: 3.83

Relevant Coursework: Probability, Statistical Programming, Bootstrapping, Regression and Inference, Numerical PDEs, Optimization, Operations Research/Graph Programming

B.S. Mathematics and Physics, May 2025

GPA: 3.87

Technical Skills

Programming: Python, C++, R, MATLAB, LaTeX

Packages and Software: NumPy, SciPy, Matplotlib, PETSc, Geant4, Shiny, Excel, Bash, Git

Research Projects

Finite Difference Discretizations of the Dirac Partial Differential Equation

May 2025–Aug 2026

- Implemented numerical algorithms for discretized Dirac equations using **PETSc** in **C++** and **Python** on a high-performance computing cluster for parallel processing.
- Evaluated numerical accuracy, stability, convergence, and computational performance to validate model implementations.

Finite Difference Schemes; Senior and Honors Capstone

Jan 2025–May 2025

- Implemented 6 finite-difference numerical schemes for partial differential equations, including wave and heat equations, using **MATLAB**.
- Analyzed explicit and implicit discretization schemes through consistency, stability, convergence, and error analysis.
- Applied Fourier and von Neumann analysis to derive numerical stability conditions and evaluate information delivery for finite-difference schemes.

Monte Carlo Simulations for Water Cherenkov Detector

Jan 2025–May 2025

- Designed and implemented Monte Carlo simulations in **C++** and **Geant4** to model particle interactions and physical detector behavior using job scheduling for HPC, including 1M+ simulations.
- Compared simulated outputs against experimental measurements and environmental datasets to assess model accuracy and validate computational results.
- Developed computational experiments and analyzed simulation behavior under varying model parameters.

Professional Experience

Graduate Administrative Assistant

Jul 2025–Aug 2026

SDSU Housing and Residential Life

- Developed and maintained data collection and reporting systems for 5,000+ operational records per semester.
- Analyzed operational trends and communicated quantitative findings to leadership teams to support staffing and resource-allocation decisions.

Undergraduate Materials Research Assistant

Aug 2024–Dec 2024

SDSU Department of Physics

- Designed experiments to synthesize and measure magnetic/thermoelectric properties of 5 novel alloys based on literature simulations
- Collaborated with faculty researchers to evaluate computational predictions.
- Presented technical findings, related them to current publications, and maintained project documentation.

Minor Projects and Additional Experience

Automatic and numeric differentiation for physics-informed neural networks

Aug 2025–Dec 2025

Line Search methods for machine learning empirical risk minimization

Jan 2025–May 2025

SDSU Community Assistant

Aug 2022–May 2025

SDSU Physics Tutor

Aug 2021–May 2025